LAB EXPERIMENT RECORD

Set 21: Energy_Consumption_Data

Department	CSE-DS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Dataset: Energy_Consumption_Data.csv

Dataset Description

The dataset contains energy consumption information for different sectors and regions, including temperature, units consumed, cost, renewable energy usage, and peak hours.

Sector	Region	Month	Temperature (°C)	Units Consumed (kWh)	Cost (₹)	Renewable Usage (%)	Peak Hours
Residential	North	Jan	15	320	2100	22	4
Commercial	South	Jan	24	540	3600	18	6
Industrial	West	Feb	20	880	5900	12	8
Residential	East	Feb	18	350	2300	25	5
Commercial	North	Mar	28	610	4100	20	7
Industrial	South	Mar	30	920	6200	15	9

1. Histogram and Density Plot for Units Consumed

Aim

To import the energy consumption dataset into R and create a histogram and density plot for Units Consumed to compare energy consumption patterns.

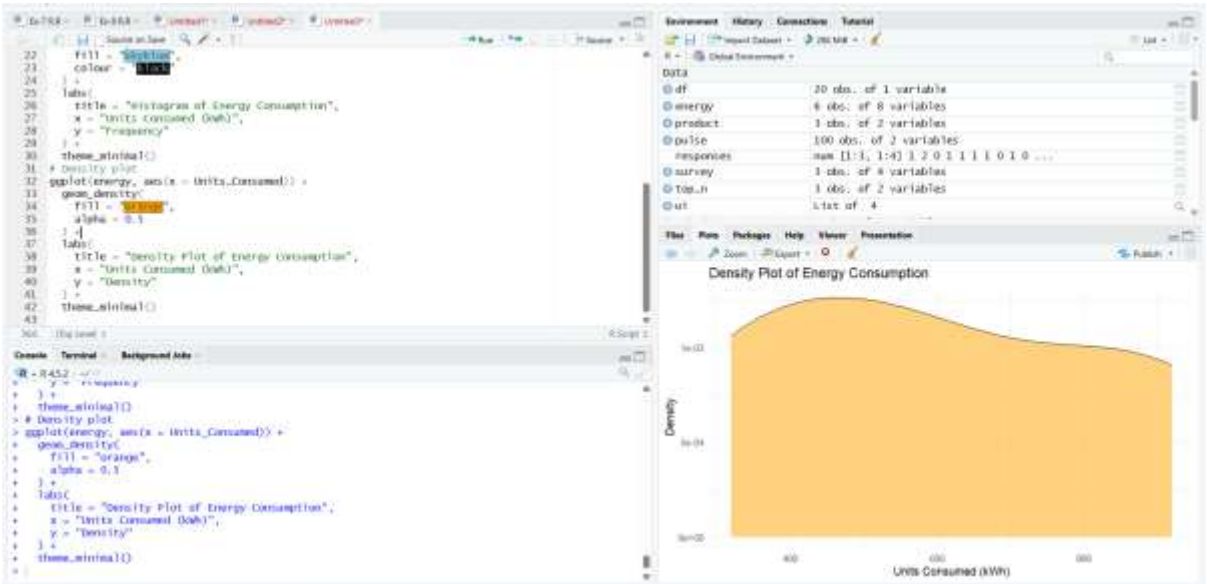
R Code

```
library(ggplot2)

energy <- data.frame(
  Sector = c(
    "Residential", "Commercial", "Industrial",
    "Residential", "Commercial", "Industrial"
  ),
  Region = c(
    "North", "South", "West",
    "East", "North", "South"
  ),
  Month = c("Jan", "Jan", "Feb", "Feb", "Mar", "Mar"),
  Temperature = c(15, 24, 20, 18, 28, 30),
  Units_Consumed = c(320, 540, 880, 350, 610, 920),
  Cost = c(2100, 3600, 5900, 2300, 4100, 6200),
  Renewable_Usage = c(22, 18, 12, 25, 20, 15),
  Peak_Hours = c(4, 6, 8, 5, 7, 9)
)

# Histogram
ggplot(energy, aes(x = Units_Consumed)) +
  geom_histogram(
    bins = 5,
    fill = "skyblue",
    colour = "black"
  ) +
  labs(
    title = "Histogram of Energy Consumption",
    x = "Units Consumed (kWh)",
```

```
  y = "Frequency"
 ) +
 theme_minimal()
# Density plot
ggplot(energy, aes(x = Units_Consumed)) +
 geom_density(
  fill = "orange",
  alpha = 0.5
 ) +
 labs(
  title = "Density Plot of Energy Consumption",
  x = "Units Consumed (kWh)",
  y = "Density"
 ) +
 theme_minimal()
```



Observation

The histogram displays the frequency of energy consumption values in different intervals, while the density plot provides a smoothed representation of their distribution. Industrial consumption is higher than residential and commercial consumption in the given dataset.

Result

The histogram and density plot were created successfully to visualize and compare energy consumption patterns.

2. Scatter Plot Between Temperature and Units Consumed

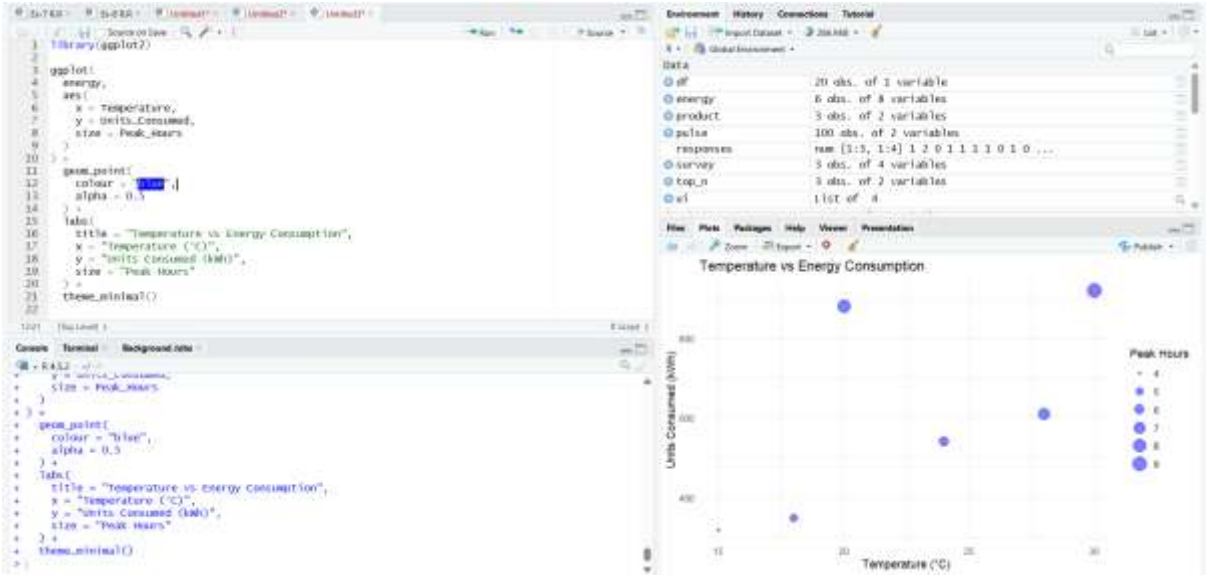
Aim

To create a scatter plot between Temperature and Units_Consumed, represent Peak_Hours using bubble size, and apply transparency to improve visualization.

R Code

```
library(ggplot2)


ggplot(
 energy,
 aes(
  x = Temperature,
  y = Units_Consumed,
  size = Peak_Hours
 )
) +
 geom_point(
  colour = "blue",
  alpha = 0.5
 ) +
 labs(
  title = "Temperature vs Energy Consumption",
  x = "Temperature (°C)",
  y = "Units Consumed (kWh)",
  size = "Peak Hours"
 ) +
 theme_minimal()
```



Observation

The scatter plot compares temperature with energy consumption. Bubble size represents peak hours, and transparency makes overlapping points easier to distinguish. In the given dataset, higher temperatures are generally associated with higher energy consumption, although the dataset is too small to establish a reliable relationship.

Result

The scatter plot was created successfully to visualize the relationship between temperature, energy consumption, and peak hours.

3. Average Renewable Usage by Sector

Aim

To calculate the average Renewable_Usage for each sector and represent the results using a bar chart.

R Code

```
library(ggplot2)

avg_renewable <- aggregate(

  Renewable_Usage ~ Sector,

  data = energy,

  FUN = mean

)

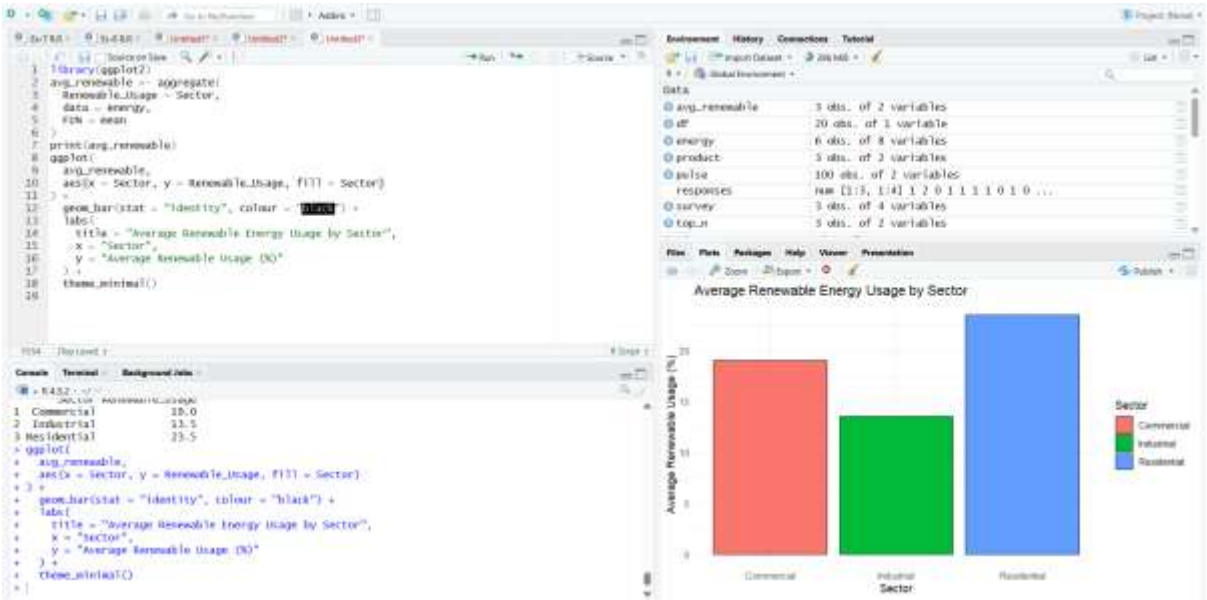
print(avg_renewable)

ggplot(

  avg_renewable,

  aes(x = Sector, y = Renewable_Usage, fill = Sector)
```

```
) +
 geom_bar(stat = "identity", colour = "black") +
 labs(
  title = "Average Renewable Energy Usage by Sector",
  x = "Sector",
  y = "Average Renewable Usage (%)"
 ) +
 theme_minimal()
```



Observation

The Residential sector has the highest average renewable energy usage at 23.5%, followed by the Commercial sector at 19%. The Industrial sector has the lowest average at 13.5%. This indicates that renewable energy adoption is comparatively higher in the Residential sector in the given dataset.

Result

The average renewable energy usage for each sector was calculated and displayed using a bar chart.

4. Tableau Dashboard for Energy Consumption

Aim

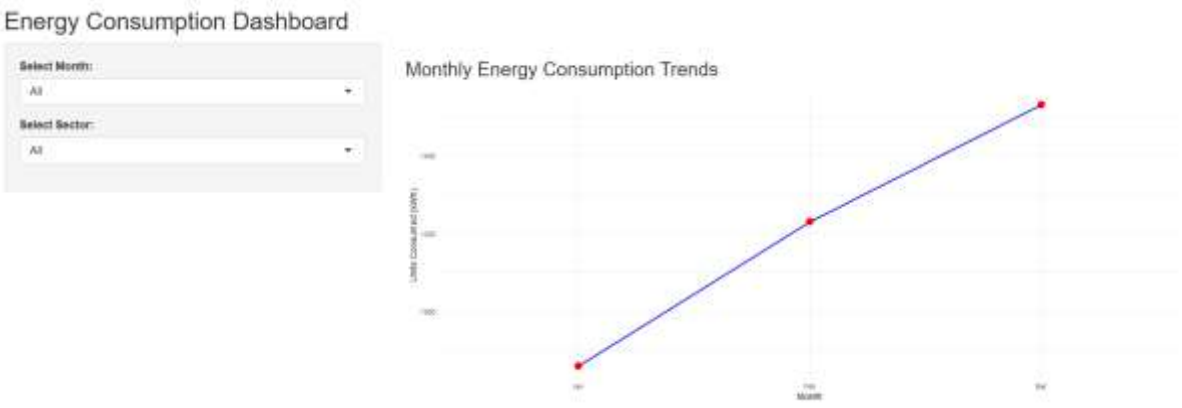
To create an interactive Tableau dashboard showing monthly energy consumption trends and a heatmap comparing sectors, regions, and units consumed, with filters for Month and Sector.

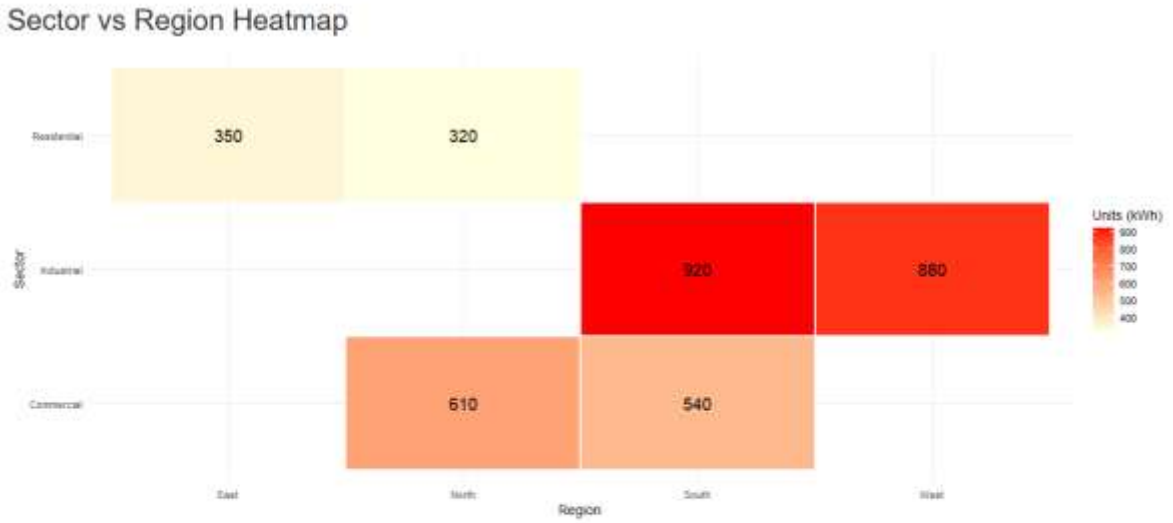
Tableau Procedure

1. Open Tableau Desktop or Tableau Public.

2. Connect to the Energy_Consumption_Data.csv file using the Text File option.
3. Open a new worksheet and rename it **Monthly Energy Trends**.
4. Drag Month to Columns and Units_Consumed to Rows.
5. Select Line Chart from the Marks or Show Me options to display monthly energy usage trends.
6. Create another worksheet named **Sector-Region Heatmap**.
7. Drag Sector to Rows and Region to Columns.
8. Drag Units_Consumed to Colour and Label in the Marks card.
9. Select Square marks and adjust the size to create a heatmap.
10. Add the Month field as a filter and select Show Filter.
11. Add the Sector field as a filter and select Show Filter.
12. Select Dashboard → New Dashboard.
13. Drag both worksheets into the dashboard and arrange them neatly.
14. Display the Month and Sector filters so users can interact with the charts.
15. Add a suitable dashboard title and save the workbook.

Output:





Filtered Energy Data

Sector	Region	Month	Temperature	Units_Consumed	Cost	Renewable_Usage	Peak_Hours
Residential	North	Jan	15.00	320.00	2100.00	22.00	4.00
Commercial	South	Jan	24.00	540.00	3600.00	18.00	6.00
Industrial	West	Feb	20.00	880.00	5900.00	12.00	8.00
Residential	East	Feb	18.00	350.00	2300.00	25.00	5.00
Commercial	North	Mar	28.00	610.00	4100.00	20.00	7.00
Industrial	South	Mar	30.00	920.00	6200.00	15.00	9.00

Observation

The line chart shows monthly energy consumption trends, while the heatmap compares energy usage across sectors and regions. The Month and Sector filters allow users to explore selected portions of the dataset.